A PROPER FLAT SPACE FINITELY PRESENTED OVER A DENSE OPEN EXTENDS TO FINITE PRESENTATION IN CODIMENSION CONTROLLED BY THE BOUNDARY

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 8624-8631 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 8633-8646, the complete author proof (12 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. One of the chapter's finite-extension family; distinct from the same-chapter flat-finite-presentation dimension cards collected earlier (this is a properness-driven finiteness propagation, no dimension count).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Openness of the quasi-finite locus of a proper morphism of spaces, the proper-plus-quasi-finite-implies-finite factorization, closedness of the image of the complement locus under finite presentation, and density bookkeeping for the chosen open.

Source reference. Original lemma 0B48: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8646–8646. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0878: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8591–8599. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04NV: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8646–8646. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0418: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8646–8646. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is flat and proper. Let $V \to Y$ be an open subspace with $|V| \subset |Y|$ dense such that $X_V \to V$ is finite. If also either f is locally of finite presentation or $V \to Y$ is quasi-compact, then f is finite.*

Proof. By Lemma 0878 the fibres of f have dimension zero. By Morphisms of Spaces, Lemma 04NV this implies that f is locally quasi-finite. By Morphisms of Spaces, Lemma 0418 this implies that f is representable. We can check whether f is finite étale locally on Y, hence we may assume Y is a scheme. Since f is representable, we reduce to the case of schemes which is More on Flatness, Lemma 0B48. □